Emotion-Aware Learning Assistant (Student Emotion Detection System)

Complete Project Documentation Set

Ideation Phase • Project Design Phase • Project Planning Phase • Testing

Ideation Phase

1. Empathy Map Canvas

Date	15 July 2026
Project Name	Emotion-Aware Learning Assistant (Student Emotion Detection System)
Maximum Marks	4 Marks

User Persona: A student engaged in self-paced or classroom learning, interacting with course material across subjects such as Computer Science, Mathematics, and Physics.

SAYS	THINKS	DOES	FEELS
"I don't get this at all." "Why does this keep failing?" "This is easy, what's next?"	"Am I the only one who doesn't understand this?" "Maybe I should just skip ahead." "I wonder how this connects to real projects."	Re-reads the same paragraph repeatedly. Skips ahead or zones out during familiar material. Searches for extra explanations online.	Frustrated when stuck on a problem for too long. Bored when material feels repetitive. Confident after solving something independently.

Pain Points

- No one notices when a student is silently confused, bored, or frustrated during self-paced learning.
- Generic, one-size-fits-all learning content doesn't adapt to how a student is actually feeling.
- Students often don't ask for help even when they need it, due to hesitation or lack of a channel to express confusion.

Gains

- Immediate, personalized guidance the moment frustration or confusion is detected.
- Encouragement and appropriately-paced content when a student is bored or already confident.
- A private, judgment-free way to express difficulty with a topic.

Ideation Phase

2. Define the Problem Statements

Date	15 July 2026
Project Name	Emotion-Aware Learning Assistant (Student Emotion Detection System)
Maximum Marks	2 Marks

PS No.	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	a student learning a new technical concept	understand a difficult topic on my own	I keep getting stuck	there's no one available to notice I'm struggling or explain it differently	frustrated and reluctant to keep trying
PS-2	a student who has already mastered a topic	stay engaged with the course	the material keeps repeating what I already know	the content isn't adapted to my current understanding	bored and disengaged from learning

Ideation Phase

3. Brainstorm & Idea Prioritization Template

Date	15 July 2026
Project Name	Emotion-Aware Learning Assistant (Student Emotion Detection System)
Maximum Marks	4 Marks

Step 1: Team Gathering & Problem Statement Selection

The team reviewed both problem statements (PS-1: struggling students go unnoticed; PS-2: advanced students disengage from repetitive content) and selected a combined focus: detecting a student's real-time emotional state from free text and responding appropriately, covering both under-support (Confused/Frustrated) and under-stimulation (Bored/Confident) cases.

Step 2: Brainstorm, Idea Listing & Grouping

- Group A — Detection approach: sentiment analysis API, custom-trained classifier, keyword/rule-based detection, facial-expression analysis (rejected — privacy-invasive and requires camera access).
- Group B — Model type: single classical ML model, deep learning (BiLSTM), transformer (BERT), ensemble of multiple models.
- Group C — Response mechanism: static canned responses only, LLM-generated personalized responses, hybrid (AI-first with template fallback).
- Group D — Delivery: browser extension, chatbot plug-in, standalone web application.

Step 3: Idea Prioritization

Idea	Priority	Rationale
Dual-model classifier (BiLSTM + BERT)	High	Higher accuracy and cross-verification vs. a single model
Hybrid AI + template response	High	Personalized guidance with reliable fallback if AI is unavailable
Standalone Streamlit web application	High	Fastest path to a usable, demonstrable product
Facial-expression analysis	Rejected	Privacy concerns and hardware dependency

Project Design Phase

4. Problem – Solution Fit Template

Date	15 July 2026
Project Name	Emotion-Aware Learning Assistant (Student Emotion Detection System)
Maximum Marks	2 Marks

Customer Segment(s)	Students in self-paced or blended learning environments across academic fields (Computer Science, Mathematics, Physics, and others).
Problem / Trigger	A student becomes confused, frustrated, bored, or (positively) confident/curious while working through material, but has no immediate, adaptive channel to express this or receive tailored help.
Emotional Response	Frustration and disengagement build silently; the student either disengages (Bored) or gives up (Frustrated) before seeking help.
Existing Alternatives	Generic FAQ pages, static help documentation, waiting for scheduled instructor office hours, or general-purpose chatbots with no emotional awareness.
Root Cause	Learning platforms treat all input the same regardless of the learner's emotional/cognitive state.
Proposed Solution	A dual-model (BiLSTM + BERT) emotion classifier detects the student's state from free-text input in real time, and an AI guidance engine (with multi-provider fallback) generates a response tailored to both the detected emotion and the academic field.
Trigger to Act	The student types a description of their problem/challenge and submits it — no separate emotion-logging step required.

Project Design Phase

5. Proposed Solution Template

Date	15 July 2026
Project Name	Emotion-Aware Learning Assistant (Student Emotion Detection System)
Maximum Marks	2 Marks

Problem Statement	Students' emotional states (confusion, frustration, boredom, curiosity, confidence) go undetected during self-paced learning, resulting in either unresolved struggle or disengagement.
Idea / Solution Description	A web application where a student describes their learning challenge in free text. Two independently trained models (a BiLSTM neural network and a fine-tuned BERT transformer) classify the emotion behind the text. An AI engine (Gemini, with OpenRouter/DeepSeek/Grok fallback) then generates a personalized, field-aware response, with a template-based fallback if all AI providers are unavailable.
Novelty / Uniqueness	Combines two complementary classification architectures with keyword-based confidence boosting and mixed-emotion detection (identifying secondary emotions above a 15% threshold), rather than relying on a single model or a single dominant label.
Social Impact / Customer Satisfaction	Gives students immediate, judgment-free support exactly when they need it, without waiting for instructor availability, potentially reducing learning drop-off caused by unaddressed frustration or disengagement.
Business Model (Revenue Model)	Freemium SaaS: free tier with template-based responses and limited AI queries per day; paid tier for institutions offering unlimited AI-generated guidance, analytics dashboards, and LMS integration.
Scalability of the Solution	Model inference is lightweight enough to run on modest hardware; the AI response layer scales horizontally across multiple provider APIs; CSV-based logging can be swapped for a proper database (e.g. PostgreSQL) as usage grows.

Project Design Phase

6. Solution Architecture

Date	15 July 2026
Project Name	Emotion-Aware Learning Assistant (Student Emotion Detection System)
Maximum Marks	4 Marks

The solution follows a layered architecture: a presentation layer (Streamlit UI), a model inference layer (BiLSTM and BERT classifiers running in parallel), an enhancement layer (keyword boosting and mixed-emotion detection), an AI orchestration layer (multi-provider fallback chain), and a persistence layer (CSV-based interaction logging).

Architecture Flow

- 1. User Input — student selects academic field and describes their problem via the Streamlit interface.
- 2. Preprocessing — text is cleaned, tokenized, and padded (BiLSTM path) or tokenized via the BERT tokenizer (transformer path).
- 3. Dual Model Inference — both the BiLSTM (~4.7M parameters) and fine-tuned BERT (bert-base-uncased) models independently produce a 5-class probability distribution.
- 4. Keyword Enhancement — explicit emotion-indicating keywords boost/suppress class probabilities, followed by renormalization.
- 5. Mixed-Emotion Detection — any secondary emotion scoring ≥ 15% is surfaced alongside the primary prediction.
- 6. AI Guidance Generation — field, problem text, detected emotion, and confidence are assembled into a prompt sent to Gemini 2.5 Flash, falling back automatically to OpenRouter, DeepSeek, then Grok, then a static template if all providers fail.
- 7. Persistence — the interaction (text, emotion, confidence, response, field, timestamp) is appended to emotion_response_examples.csv; new emotion-response pairs are logged to emotion_response_mapping.csv.
- 8. Analytics — session history feeds a three-tab Plotly dashboard (Emotions, Fields, Summary).

Figure 1: High-Level Architecture Diagram (Description)

[Insert architecture diagram] Student Input → Streamlit UI → {BiLSTM Model, BERT Model} → Keyword Enhancement + Mixed-Emotion Detection → AI Guidance Engine (Gemini → OpenRouter → DeepSeek → Grok → Template Fallback) → Response Display + CSV Persistence → Analytics Dashboard.

7. Solution Requirements (Functional & Non-Functional)

Date	15 July 2026
Project Name	Emotion-Aware Learning Assistant (Student Emotion Detection System)
Maximum Marks	4 Marks

Functional Requirements

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Field & Problem Input Capture	Field selection dropdown (11 subjects) • Free-text problem description • Quick-example buttons
FR-2	Emotion Detection	BiLSTM prediction • BERT prediction • Keyword-based confidence boosting • Mixed-emotion detection (≥15% threshold)
FR-3	AI-Powered Response Generation	Gemini prompt construction • Multi-provider fallback (OpenRouter, DeepSeek, Grok) • Template fallback when AI unavailable/disabled
FR-4	Session & Data Persistence	Session history tracking (per-model entries) • CSV logging of every interaction • Emotion-response mapping updates
FR-5	Analytics Dashboard	Emotion distribution pie chart • Confidence timeline • Field-based breakdown • Summary statistics
FR-6	User Controls	AI response toggle • CSV save toggle • Analysis detail visibility toggle • Clear history button

Non-Functional Requirements

NFR No.	Requirement	Description
NFR-1	Usability	Single-page interface with clear field/problem inputs, quick-example buttons, and plain-language guidance responses requiring no technical knowledge.
NFR-2	Security	API keys stored in environment variables (.env, excluded from version control), never exposed client-side.
NFR-3	Reliability	Four-provider AI fallback chain (Gemini → OpenRouter → DeepSeek → Grok) plus a static template fallback ensures a response is always returned.
NFR-4	Performance	Models cached via Streamlit's cache_resource, loaded once per server session; BERT inference accelerated via CUDA GPU where available.
NFR-5	Availability	Application designed to degrade gracefully — BERT-only mode if BiLSTM fails to load, template responses if all AI providers fail.
NFR-6	Scalability	CSV-based logging can be replaced with a relational database as data volume grows; model inference is stateless and horizontally scalable.

8. Data Flow Diagram & User Stories

Date	15 July 2026
Project Name	Emotion-Aware Learning Assistant (Student Emotion Detection System)
Maximum Marks	4 Marks

Data Flow Diagram (Description)

[Insert DFD] Student (external entity) → [Process: Capture Field + Problem] → [Process: Preprocess Text] → [Process: Dual-Model Emotion Classification] → [Data Store: Model Weights] → [Process: Keyword Enhancement & Mixed-Emotion Detection] → [Process: AI Response Generation] → [Data Store: emotion_response_examples.csv / emotion_response_mapping.csv] → [Process: Render Response & Analytics] → Student.

User Stories

User Type	Epic	Story No.	User Story / Task	Acceptance Criteria	Priority
Student	Input Capture	USN-1	As a student, I can select my field of study and describe my problem in free text.	Field dropdown and text area both render and accept input	High
Student	Emotion Detection	USN-2	As a student, I want the system to detect my emotional state from what I write.	Both BiLSTM and BERT return an emotion label with confidence	High
Student	Emotion Detection	USN-3	As a student, I want to see if I'm feeling more than one emotion at once.	Mixed emotions display when a secondary score ≥ 15%	Medium
Student	AI Guidance	USN-4	As a student, I want a personalized, supportive response based on my detected emotion.	AI response references field, emotion, and offers a concrete tip	High
Student	AI Guidance	USN-5	As a student, I want a response even if the AI service is down.	Template fallback returns a valid response when all AI providers fail	High
Student	Personalization	USN-6	As a student, I can toggle AI responses off in favor of instant template responses.	Unchecking the AI toggle returns a template response with no API call	Medium
Student	Analytics	USN-7	As a student, I want to see how my emotions have changed across my session.	Emotions tab shows a distribution chart and a confidence timeline	Low
Administrator	Data Logging	USN-8	As an administrator, I want every interaction logged for future model improvement.	Each submission appends a row to emotion_response_examples.csv	Medium

9. Technology Stack (Architecture & Stack)

Date	15 July 2026
Project Name	Emotion-Aware Learning Assistant (Student Emotion Detection System)
Maximum Marks	4 Marks

Table 1: Components & Technologies

Component	Description	Technology
User Interface	How the student interacts with the application	Streamlit (Python-based web UI)
Application Logic-1	Sequence classification (fast baseline model)	TensorFlow / Keras — Bidirectional LSTM
Application Logic-2	Transformer-based classification (higher accuracy)	PyTorch + Hugging Face Transformers — fine-tuned bert-base-uncased
Application Logic-3	Text preprocessing & tokenization	NLTK, Keras Tokenizer, Hugging Face BertTokenizerFast
External API-1	Primary AI response generation	Google Gemini 2.5 Flash API
External API-2/3/4	Fallback AI response generation	OpenRouter, DeepSeek, Grok (xAI) — OpenAI-compatible APIs
Machine Learning Model	5-class emotion classification (Bored, Confident, Confused, Curious, Frustrated)	BiLSTM (~4.7M params) and fine-tuned BERT, both domain-adapted
Data Storage	Interaction logging and emotion-response mapping	CSV files (emotion_response_examples.csv, emotion_response_mapping.csv)
Model Training Infrastructure	GPU-accelerated model training	Kaggle dual-GPU (T4 x2) notebook environment
Infrastructure (Local)	Local inference and app hosting	Python 3.11 virtual environment, NVIDIA RTX 3050 (CUDA 12.1)

Table 2: Application Characteristics

Characteristic	Description	Technology
Open-Source Frameworks	Core ML and web frameworks used	TensorFlow, Keras, PyTorch, Hugging Face Transformers, Streamlit, Plotly, NLTK, scikit-learn
Security Implementations	Protection of API credentials	python-dotenv for environment-variable isolation; .env excluded from version control
Scalable Architecture	Justification of scalability	Stateless model inference layer with cached model loading (st.cache_resource); multi-provider API fallback distributes load
Availability	Justification of availability	4-tier AI provider fallback chain plus static template fallback guarantees a response even during upstream outages
Performance	Design considerations for speed	GPU-accelerated BERT inference (CUDA); models loaded once and cached across the session

10. Project Planning Template (Product Backlog, Sprint Planning)

Date	15 July 2026
Project Name	Emotion-Aware Learning Assistant (Student Emotion Detection System)
Maximum Marks	5 Marks

Product Backlog & Sprint Schedule

Sprint	Epic	Story No.	User Story / Task	Story Points	Priority	Status
Sprint-1	Environment & Data Setup	USN-1	Configure Kaggle GPU environment, install pinned dependencies, verify dataset availability	3	High	Done
Sprint-1	Data Preprocessing	USN-2	Source, combine, and map 3 datasets to 5 target emotion classes; generate synthetic data for the missing Bored class	5	High	Done
Sprint-1	Data Preprocessing	USN-3	Text cleaning, tokenization, and sequence padding for BiLSTM input	2	High	Done
Sprint-2	Model Training	USN-4	Train baseline BiLSTM classifier with focal loss and class weighting	5	High	Done
Sprint-2	Model Training	USN-5	Fine-tune bert-base-uncased on the same dataset	5	High	Done
Sprint-2	Model Training	USN-6	Domain-adaptive fine-tuning of both models on student-context data; validate on novel hand-written sentences	3	Medium	Done
Sprint-3	Prediction Pipeline	USN-7	Build unified prediction schema, keyword enhancement, and mixed-emotion detection	3	High	Done
Sprint-3	AI Guidance Engine	USN-8	Integrate Gemini API with prompt construction; build 4-provider fallback chain and template fallback	5	High	Done
Sprint-4	Application UI	USN-9	Build Streamlit interface: field/problem input, settings panel, model comparison view	5	High	Done
Sprint-4	Persistence & Analytics	USN-10	CSV logging, session history, and 3-tab Plotly analytics dashboard	3	Medium	Done
Sprint-4	Deployment	USN-11	Reproducible environment setup (requirements.txt), cross-browser validation, end-to-end testing	2	Medium	Done

Project Tracker, Velocity & Burndown

Sprint	Total Story Points	Duration	Start Date	Planned End	Status
Sprint-1	10	6 Days	01 Jul 2026	06 Jul 2026	Completed
Sprint-2	13	6 Days	07 Jul 2026	12 Jul 2026	Completed
Sprint-3	8	4 Days	13 Jul 2026	16 Jul 2026	Completed
Sprint-4	10	5 Days	17 Jul 2026	21 Jul 2026	Completed

11. User Acceptance Testing (UAT)

Date	15 July 2026
Project Name	Emotion-Aware Learning Assistant (Student Emotion Detection System)
Maximum Marks	[Fill in]

Project Overview

Project Name: Emotion-Aware Learning Assistant (Student Emotion Detection System)

Project Description: A dual-model (BiLSTM + BERT) system that detects student emotional states from free text and generates AI-powered, field-aware learning guidance.

Project Version: 1.0

Testing Period: 10 July 2026 to 15 July 2026

Testing Scope

- Field & problem input capture, including quick-example buttons
- BiLSTM and BERT emotion prediction accuracy and confidence display
- Mixed-emotion detection and display
- AI response generation and multi-provider fallback chain
- CSV persistence and session history tracking
- Analytics dashboard (Emotions, Fields, Summary tabs)

Testing Environment

URL/Location: Local Streamlit deployment (streamlit run streamlit_app.py)

Test Cases

TC ID	Test Scenario	Test Steps	Expected Result	Actual Result	Pass/Fail
TC-001	Detect Frustrated emotion from clear input	Enter "Why does this keep failing no matter what I try" and click Get AI Learning Help	System predicts Frustrated with high confidence (>90%)	Predicted Frustrated at 99.9%	Pass
TC-002	Detect Bored emotion from novel phrasing	Enter "I already know all of this, can we move faster" (not in training templates)	System predicts Bored	Predicted Bored at 99.9% after domain adaptation	Pass
TC-003	Mixed emotion display	Enter ambiguous input scoring above 15% on two emotions	UI shows combined label e.g. "Curious + Confused"	Mixed emotion label rendered correctly	Pass
TC-004	AI fallback chain on quota exhaustion	Trigger a Gemini quota error and observe provider used	System falls through to OpenRouter/DeepSeek/Grok automatically	Fallback to next provider succeeded, logged to console	Pass
TC-005	Template fallback when AI disabled	Uncheck "Use AI Response" and submit a problem	Instant template response returned, no API call made	Template response returned instantly	Pass
TC-006	CSV persistence	Submit 3 different problems with "Save to CSV" enabled	emotion_response_examples.csv grows by 3 rows	3 new rows confirmed in CSV	Pass

TC-007	Cached model loading	Submit multiple predictions in one session	Models load once; response time stays consistent across repeated interactions	Models cached via st.cache_resource; no reload observed	Pass
TC-008	Analytics dashboard rendering	Submit several interactions, open all three analytics tabs	Emotions, Fields, and Summary tabs all render correctly with charts	All three tabs rendered without error	Pass
TC-009	Cross-browser consistency	Open the application in two different browsers	Consistent layout, widgets, and visualizations in both	Consistent rendering confirmed	Pass

Bug Tracking

Bug ID	Description	Steps to Reproduce	Severity	Status	Notes
BG-001	BiLSTM predictions flattened to near-uniform confidence (~15-40%)	Run bilstm_predictor.py against any test sentence	High	Closed	Root cause: redundant manual softmax applied on top of the model's built-in softmax output layer
BG-002	Keras 3.13.2 model fails to load locally (module version mismatch)	Load .keras file saved from Kaggle in a Python 3.10 local environment	High	Closed	Resolved by creating a dedicated Python 3.11 virtual environment matching the Kaggle Keras version
BG-003	TensorFlow DLL load failure on Windows	Import tensorflow in the new Python 3.11 venv	Medium	Closed	Resolved by installing Microsoft Visual C++ Redistributable and reordering imports

BG-004	Gemini API returned HTTP 429 quota-exceeded error	Call get_gemini_response repeatedly under free-tier limits	Medium	Closed	Resolved by implementing the 4-provider fallback chain